

Day - 10

The Controlled-NOT (CNOT) Gate

Until today, all quantum operations we covered (X, Y, Z, H) acted on a single qubit. However, real computational power comes from qubits interacting with one another. The **Controlled-NOT (CNOT)** gate is our first two-qubit gate and is the fundamental building block for **Quantum Entanglement**.

1. Anatomy of the CNOT Gate

The CNOT gate acts on two distinct qubits:

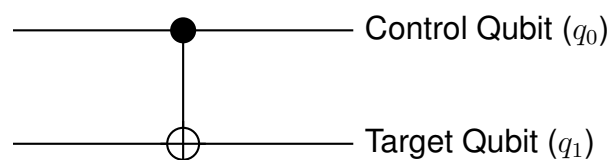
- **Control Qubit:** Determines whether an action should occur. (*Remains unchanged after the operation*).
- **Target Qubit:** The qubit that receives the conditional operation.

The Golden Rule of CNOT

- If the control qubit is $|0\rangle$, the target qubit does nothing.
- If the control qubit is $|1\rangle$, apply an **X (NOT)** gate to the target qubit (it flips).

2. Visualizing the Circuit

In quantum circuit diagrams, the CNOT gate is represented by a solid dot on the control wire, connected by a vertical line to a crossed circle ($\oplus$) on the target wire.



3. Mathematical Representation & Truth Table

Because the CNOT gate acts on two qubits, it operates on a 4-dimensional state space ($|00\rangle, |01\rangle, |10\rangle, |11\rangle$). Mathematically, it is represented as a 4×4 unitary matrix:

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Input		Output	
Control	Target	Control	Target
$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$
$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$ (Target Flipped)
$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 0\rangle$ (Target Flipped)

4. Implementation in Qiskit

In IBM's Qiskit, applying a CNOT gate is done using the `cx()` method. The first argument is the control qubit, and the second is the target qubit.

Python/Qiskit Code

```
from qiskit import QuantumCircuit
# Create a circuit with 2 qubits and 2 classical bits
qc = QuantumCircuit(2, 2)

# Prepare control qubit in state |1>
qc.x(0)

# Apply CNOT: Control is qubit 0, Target is qubit 1
qc.cx(0, 1)

# Measure the results
qc.measure([0,1], [0,1])
```

5. Key Properties & Common Mistakes

- **Reversibility:** Like all quantum gates, CNOT is its own inverse. Applying CNOT twice in a row leaves the qubits exactly as they started ($CNOT^2 = I$).

- **Misconception - "Both qubits change":** False. The control qubit's state dictates the action, but its own state remains completely unaffected by the CNOT operation.
- **Misconception - "Target always flips":** False. The target only flips conditionally if the control is $|1\rangle$.